

PROFESSIONAL SUMMARY

PhD candidate in theoretical particle physics with **four** research papers (three published in Physical Review D, fourth as an arXiv preprint), **50+** citations, **eight** conference talks, and invited seminars at Brookhaven National Lab and CERN. Specializing in large-scale numerical computation, symbolic manipulation, and multi-parameter optimization applied to high-dimensional scientific datasets. Interested in foundation models for scientific data, mechanistic interpretability, and applying physics-trained quantitative methods to large-scale ML research.

EDUCATION

- **Stony Brook University** Aug 2020 – Present
Ph.D. in Physics (High Energy Theory and Phenomenology)
- **Stony Brook University** Aug 2018 – June 2020
M.A. in Physics
- **Indian Institute of Technology Roorkee** July 2014 – May 2018
B.Tech., Electronics and Communication Engineering, Minor in Physics

RESEARCH EXPERIENCE

- **Graduate Research Assistant, C.N. Yang Institute for Theoretical Physics** Aug 2020 – Present
Stony Brook University
 - Designed and implemented Monte Carlo simulation workflows generating **1M+** events for collider physics, conducting data-driven research across high-dimensional parameter spaces.
 - Identified **eight** relevant parameters, performed multi-parameter regression via minimization of chi-squared loss function, and evaluated fit quality across parameter sweeps.
 - Constructed dimension-eight basis of **175** operators for universal effective field theories filtered from **44,807**. Cross-checked using group theory and representation theory techniques to compute group invariants.
 - Collaborated in multi-institutional research teams; co-authored multiple peer-reviewed publications in Physical Review D journal.
- **Research Assistant, Center for Nuclear Science** Aug 2018 – June 2020
Stony Brook University
 - Experience with large-scale simulation packages generating **1M+** events and C++ based data analysis framework ROOT, executed via HTCondor job submission on the Brookhaven National Lab computing cluster.
 - Applied statistical reweighting techniques to estimate projected uncertainty reduction of $\sim 20\%$ for the proposed electron-ion collider.

TECHNICAL SKILLS

- **Machine Learning & Data Science:** PyTorch, Scikit-learn, Matplotlib, Pandas, Jupyter, statistical inference, high-dimensional modeling, data analysis
- **Programming:** Python, C, C++, Java, SQL
- **Mathematical Methods:** Probability theory, stochastic processes, hypothesis testing, multi-parameter regression
- **Scientific Computing:** SciPy, NumPy, Mathematica, MATLAB, ROOT, IBM Qiskit
- **High-Performance Computing:** HTCondor, Linux cluster computing
- **Simulation & Modeling:** Monte Carlo Methods: MadGraph
- **Tools:** Git, LaTeX, Claude Code

CONFERENCE/INVITED TALKS

- **Brookhaven National Lab (BNL) Seminar (invited), Brookhaven, December 11th 2025**
“Universal Standard Model Effective Field Theory beyond dimension-six”
- **Higgs and Effective Field Theory (HEFT), CERN, June 3rd 2025**
“Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory”
- **Large Hadron Collider (LHC) EFT Working Group meeting, (invited), April 28th 2025**
“Drell-Yan production in universal theories beyond dimension-six SMEFT”
- **American Physical Society (APS) DPF-Pheno 2024, University of Pittsburgh, May 13th 2024**
“Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory”
- **American Physical Society (APS) April Meeting, 2020**
“Strange quark polarized distribution function using SIDIS at EIC”

RELEVANT PROJECTS

- **Multilayer Perceptron from Scratch** CSE 353 Machine Learning
Stony Brook University
 - Implemented a multilayer perceptron from scratch in PyTorch including manual forward/backward passes, gradient derivations, and training loops; evaluated on synthetic classification datasets (linearly separable blobs and spirals).
- **Quantum Error Correction** PHY 605 Quantum Programming
Stony Brook University
 - Implemented quantum circuits in IBM Qiskit and re-implemented published IBM Quantum simulations ([arXiv:2309.02863](#)) to investigate the Nishimori phase transition in quantum error correcting codes and analyze error thresholds.

PUBLICATIONS

All papers can be found on my [INSPIRE](#) and [Google Scholar](#) profile. Authors ordered alphabetically.

- “*USMEFT as a tool for discovery of universal new physics at high luminosity LHC*”
[arXiv:2605.15262 \[hep-ph\]](#)
Tyler Corbett, **Jay Desai**, O. J. P. Eboli, M. C. Gonzalez-Garcia, Matheus Martinez
- “*Drell-Yan production in universal theories beyond dimension-six SMEFT*”
[arXiv:2503.19962 \[hep-ph\]](#), *Phys. Rev. D* **112**, 013009 (2025)
Tyler Corbett, **Jay Desai**, O. J. P. Eboli, M. C. Gonzalez-Garcia, Matheus Martinez, Peter Reimitz
- “*Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory*”
[arXiv:2404.03720 \[hep-ph\]](#), *Phys. Rev. D* **110** (2024) 033003
Tyler Corbett, **Jay Desai**, O.J.P. Eboli, M.C. Gonzalez-Garcia
- “*Impact of dimension-eight SMEFT operators in the EWPO and Triple Gauge Couplings analysis in Universal SMEFT*”
[arXiv:2304.03305 \[hep-ph\]](#), *Phys. Rev. D* **107**, 115013 (2023)
Tyler Corbett, **Jay Desai**, O.J.P. Eboli, M.C. Gonzalez-Garcia, Matheus Martinez, Peter Reimitz

AWARDS & ACHIEVEMENTS

- Gerald Brown Prize (Research Travel Award), Stony Brook University
- Nationwide Education and Scholarship Test (Ranked **13th** nationally)
- Joint Entrance Examination (JEE) Advanced (Ranked **971** nationally)

LEADERSHIP & TEACHING

- Instructor for PHY 121, PHY 122, PHY 133, PHY 134.
- Teaching Assistant for PHY 301, PHY 308, PHY 503.
- Journal Referee for the Journal of High Energy Physics.